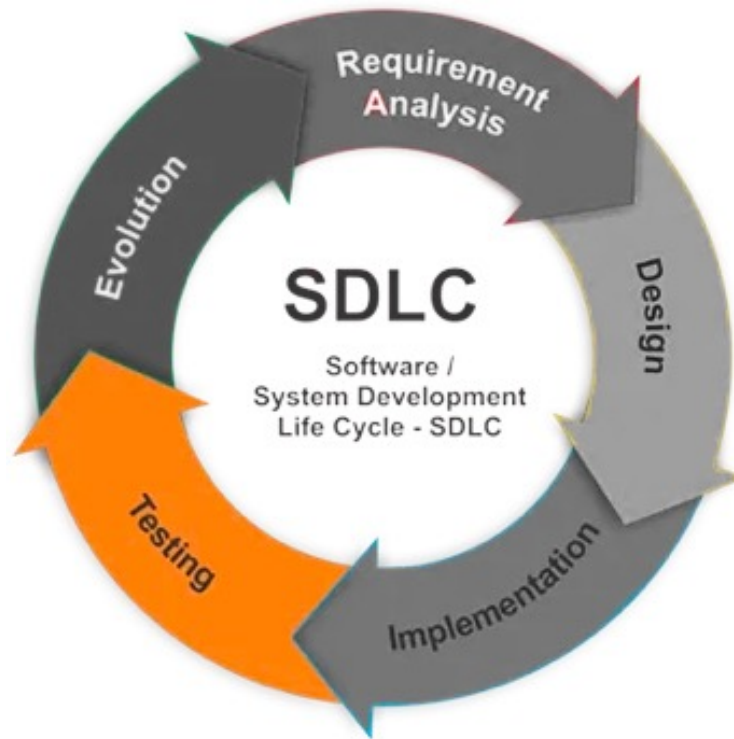




Chapter 8: Software Testing

Use of Software Testing

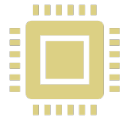
- Detect Defects and Failures
- Reduce Development Cost
- Improve Performance



Definition & Purpose of Software Testing



Software testing is an investigation conducted to provide stakeholders with information about the quality of the software product or service under test.



Software testing include a process which **evaluate the functionality of a software application** with an intent to find **whether the developed software met the specified requirements** and to identify the defects to ensure that the product is defect-free in order to produce a quality product.



Why Software testing is important?

Defects in software systems can cause a significant effect on our day-to-day lives.

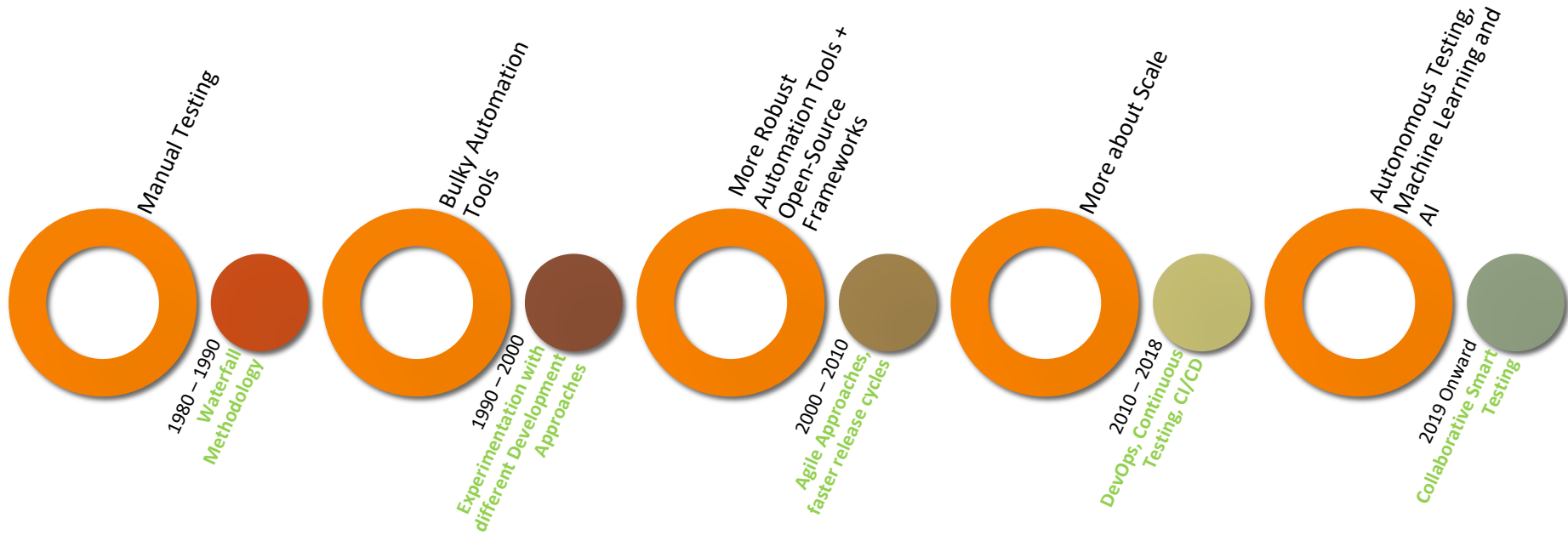
Catastrophic Event without proper software testing:

In the 1980s, software defects in the code controlling the Therac-25 radiation therapy machine were directly responsible for some patient deaths.

In 2011, Honda Company was forced to recall 2.49 million cars and minivans worldwide to repair a software problem that could damage the automatic transmission.

Such issues can cause loss of life and the reputation of a company.

Evolution of Testing

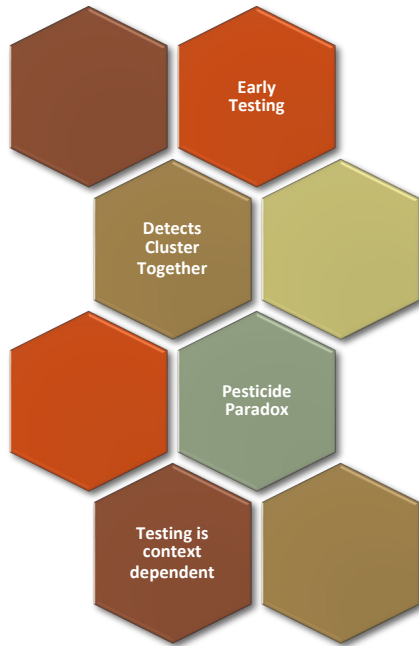


TESTING APPROACH

Principles, Methods & Framework



Principles of software testing



Early testing

- save money when issues are discovered and fixed early.

Detect cluster together

- certain components or modules of software usually contain the most no. of issues.

Pesticide paradox

- Same tests ran repeatedly will fail to find new issues. Find new effective way to test.

Testing is context dependent

- testing is all about context.

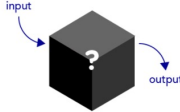
Test Methods

Black box testing - focuses on the input and output without knowing the internal implementation

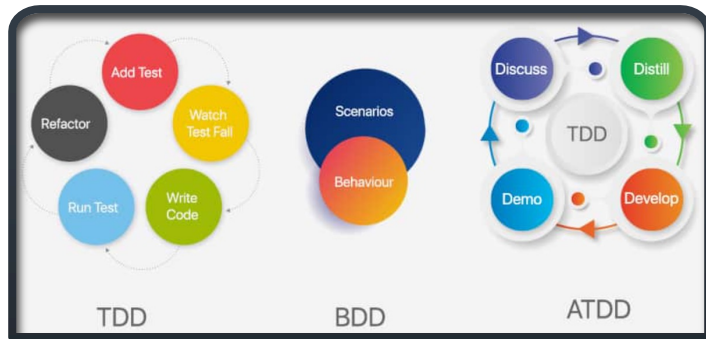
White box testing - focuses majorly on internal implementation

- Common Technique:
 - Loop Testing
 - Data Flow Testing
 - Condition Testing
 - Static Testing Methods

Grey box testing - an effective combination of Black Box Testing and White Box Testing.

Black-box Testing	White-box Testing	
		
hidden	Internal Structure	known
Not Required	Implementation knowledge	Required
Not Required	Programming knowledge	Required
Functional, behavioural test	Type of Test	Structural, logic test
Initiated by requirement specs.	Test phase	After detail design docs.
High level testing	Test Level	Low level testing

Testing framework



Test-Driven Development (TDD)

- development technique that practices of writing a test and implementation of a features and writing unit test case.

Behavioral Driven Development (BDD)

- development technique that practices of creating simple scenarios on how an application should behave.

Acceptance Test-Driven Development (ATDD)

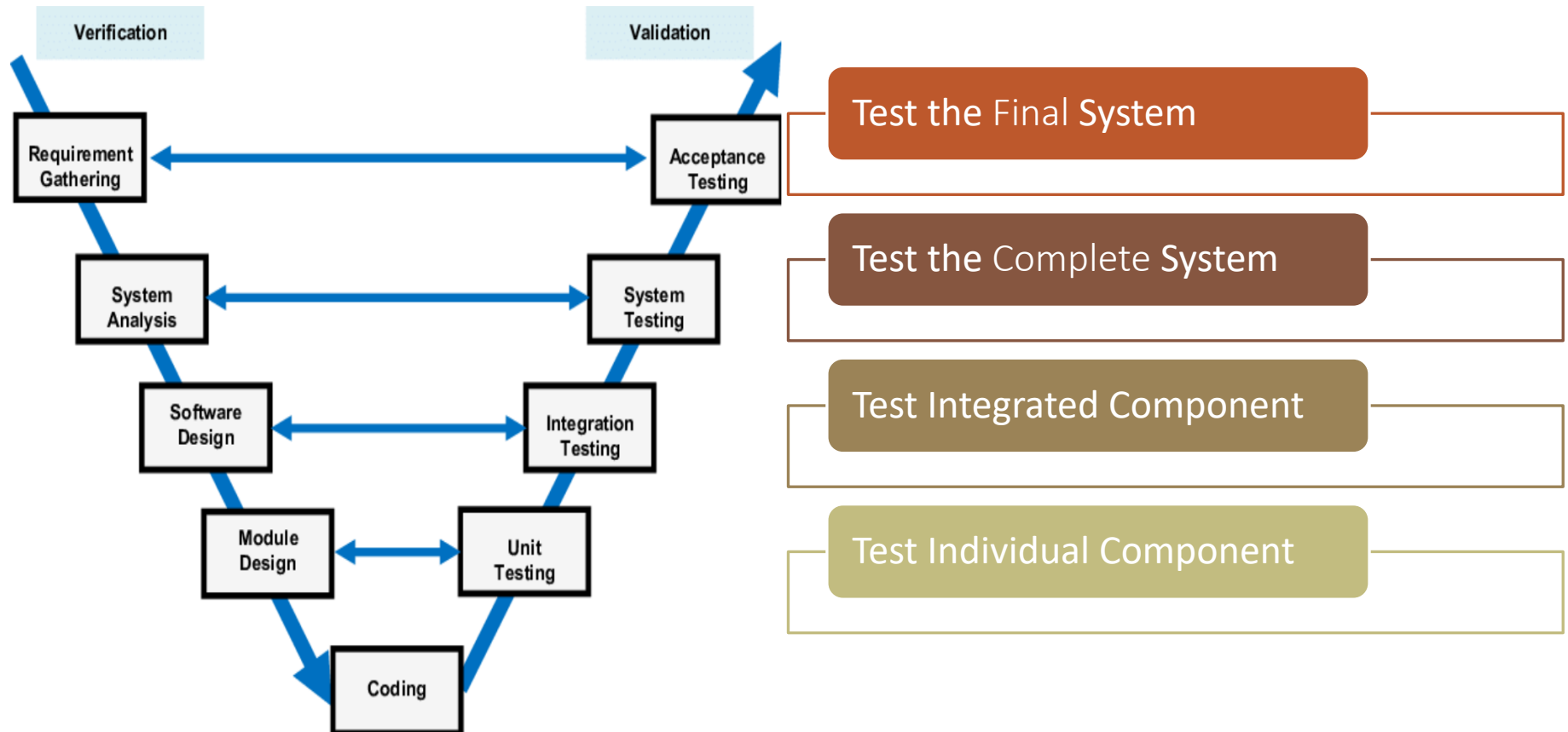
- development technique that practices of capturing user requirement criteria to tests.

	TDD	BDD	ATDD
WHAT	Focuses on the implementation of a feature	Focuses on the system's behaviour	Focuses on capturing the accurate requirement
WHO	Mainly developers involve in this to write Unit Tests	Developers, QAs & Customers involve in this process	Developers, QAs and Customers involve in this process
HOW	Focus is to write Unit Tests	Focus is on understanding requirements	Focus is to write Acceptance Tests

TESTING APPROACH

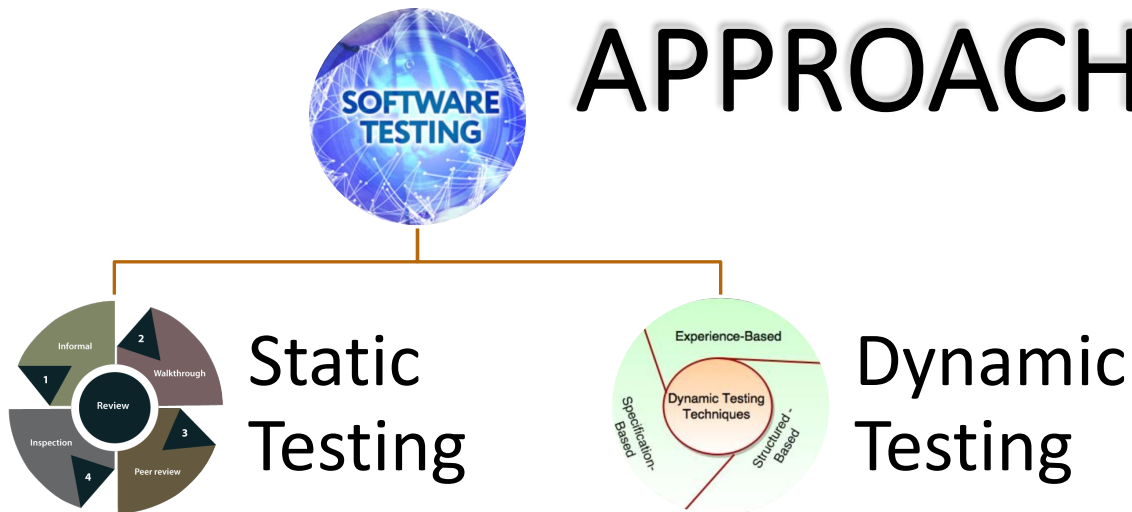
Understand Testing levels

Verification & Validation Model



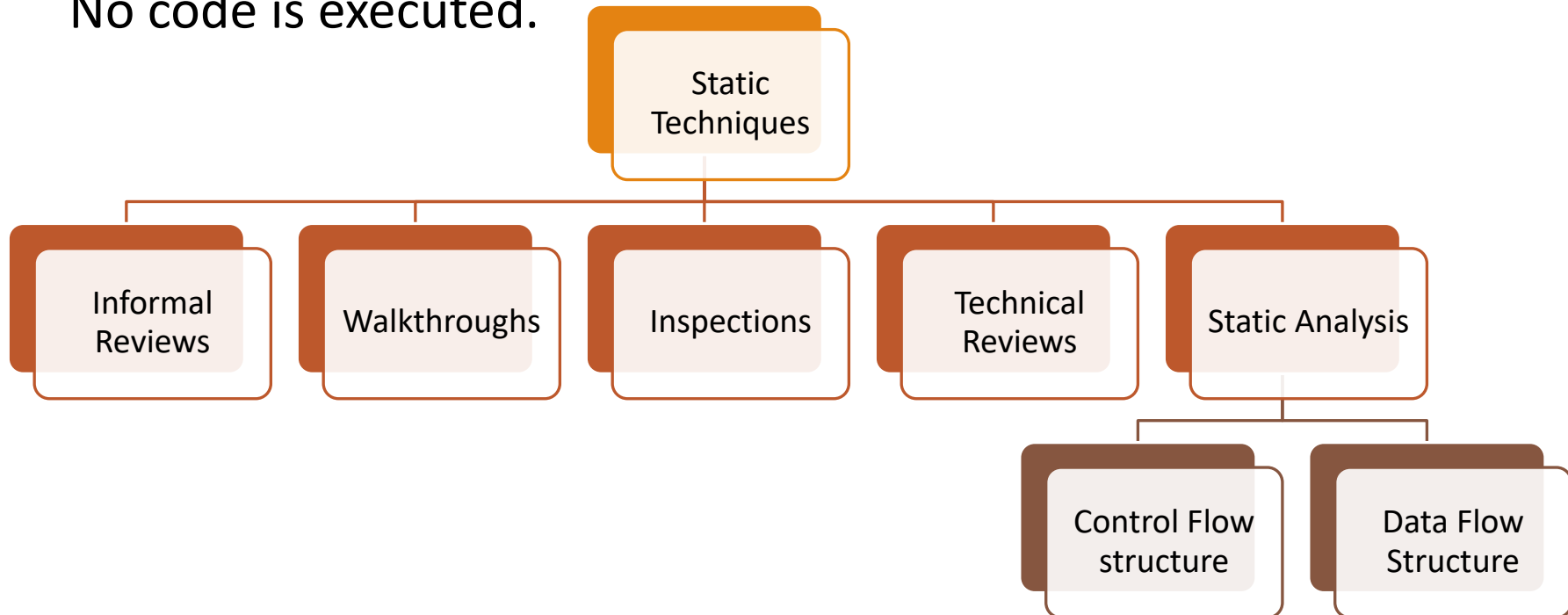
Static & Dynamic Techniques

TESTING APPROACH



Static Testing Techniques

Static testing is part of verification activities where in you have to verify that we are building the product right as per the standards set by the organization.
No code is executed.



Static Testing Techniques (2)

Informal review

- the creator of the documents put the contents in front of audience, and everyone gives their opinion and thus defects are identified in the early stage.

Walkthrough

- is basically performed by experienced person or expert to check the defects so that there might not be problem further in the development or testing phase.

Peer review

- means checking documents of one-another to detect and fix the defects. It is basically done in a team of colleagues.

Static Testing Techniques (3)

Inspection

- is basically the verification of document by higher authority like the verification of software requirement specifications.

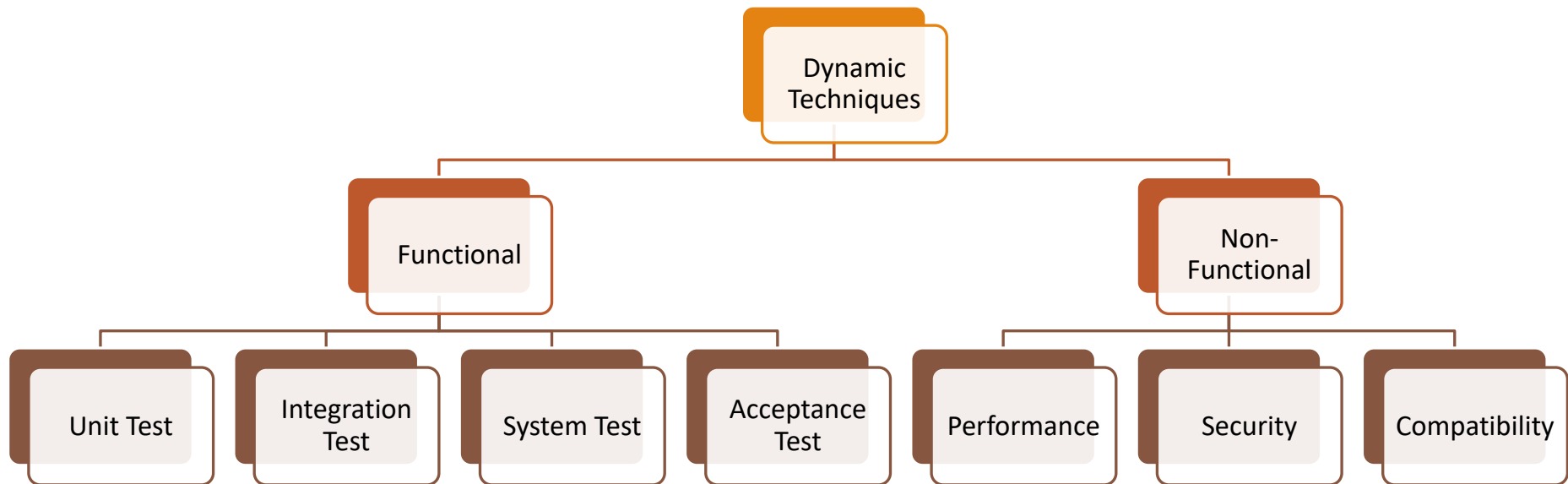
Static Analysis

- includes the evaluation of the code quality that is written by developers. Different tools are used to do the analysis of the code and comparison of the same with the standard.
 - Control flow structure
 - Data flow structure

Dynamic Testing Techniques

Dynamic testing

- is related to Validation activities where in you are giving an input and expect an output.



Dynamic Testing Techniques – Functional Testing

Functional Testing is defined as a type of testing which verifies that each function of the software application operates in conformance with the requirement specification.

- ❑ **Unit Testing** involves testing individual components of the software program or application
- ❑ **Integration Testing** is a type of software testing, which is performed on software to determine the flow between two or more modules by combining them.

Dynamic Testing Techniques – Functional Testing(2)

- ❑ **System Testing** is a level of testing that validates the complete and fully integrated software product.
 - ❑ The purpose of a system test is to evaluate the end-to-end system specifications.
- ❑ **Acceptance Testing** is a level of software testing where a system is tested for acceptability.
 - ❑ The purpose of this test is to evaluate the system's compliance with the business requirements and assess whether it is acceptable for delivery.

Dynamic Testing Techniques – Non-Functional Testing

Non-Functional Testing are used to determine the performance of the system and to validate or verify the quality attribute of the system.

□ **Performance Testing** is a software testing process used for testing the speed, response time, stability, reliability, scalability and resource usage of a software application under particular workload.

Dynamic Testing Techniques – Non-Functional Testing (2)

- ❑ **Security testing** is a type of software testing that uncovers vulnerabilities, threats, risks in a software application and prevents malicious attacks from intruders.
- ❑ **Compatibility Testing** is a type of software testing to check whether your software is able of operating on different hardware, operating systems, applications, network environments or Mobile devices.

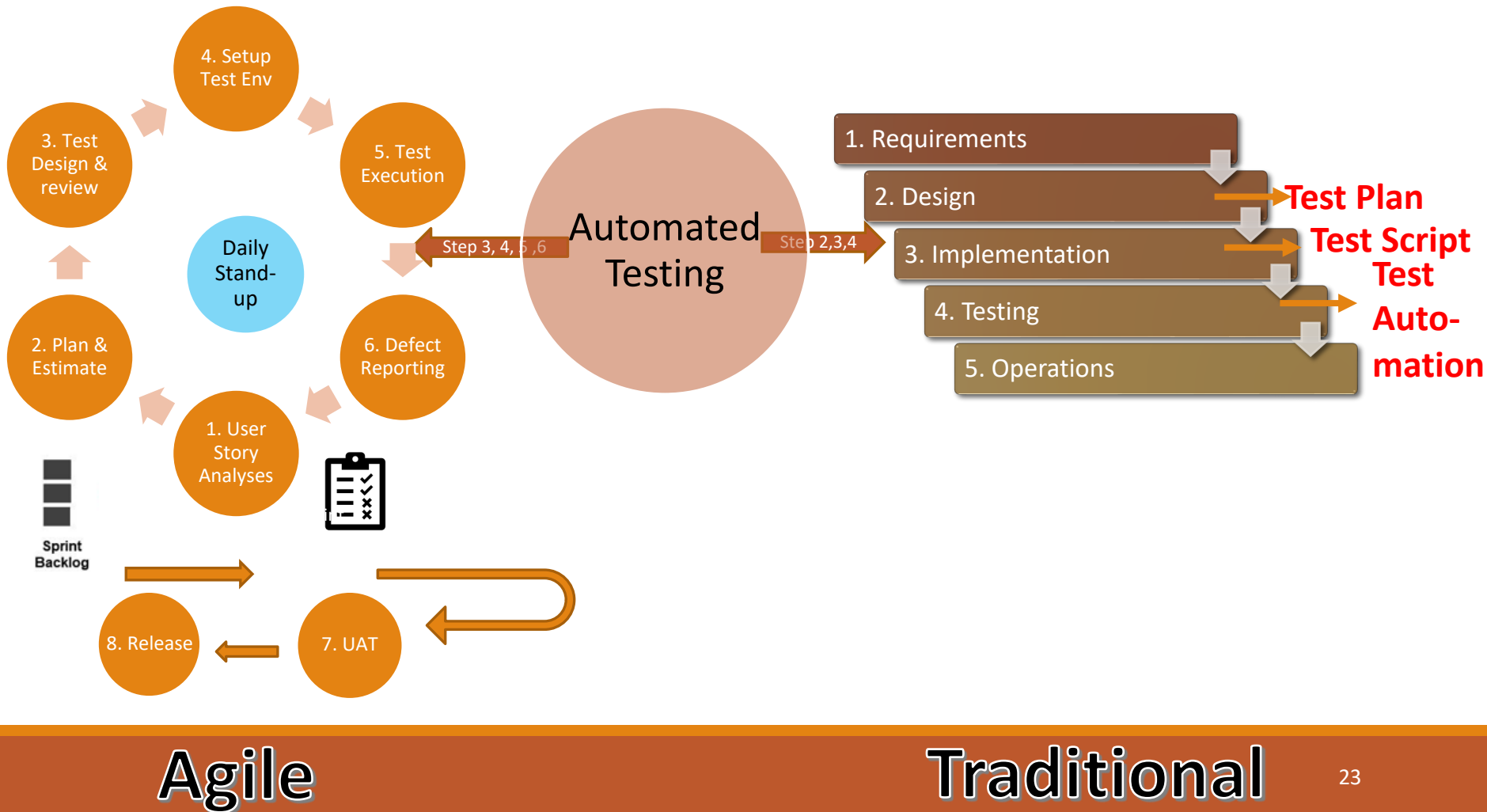
Manual & Automated Testing

*Traditional, Agile and DevOps
Methodology*

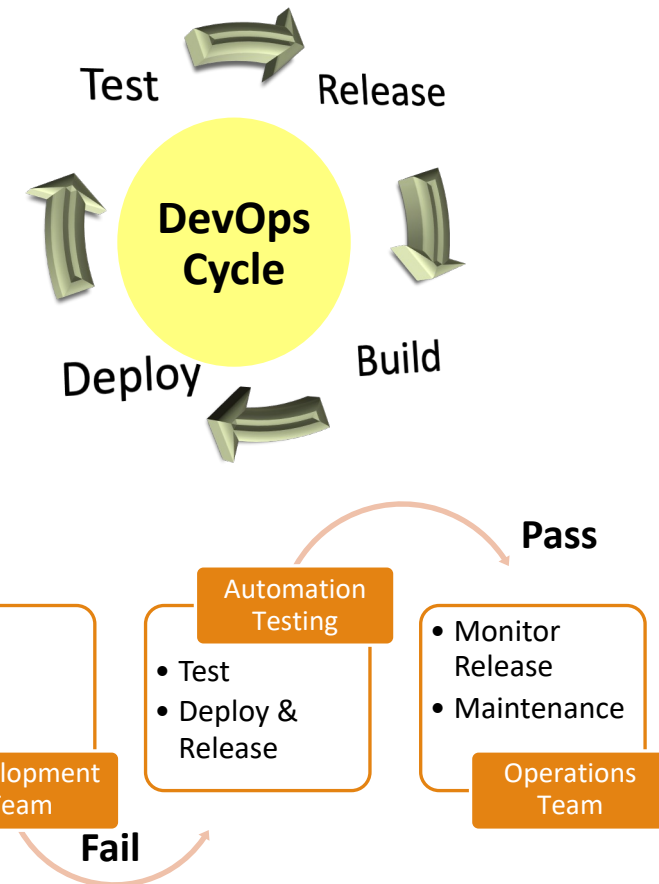
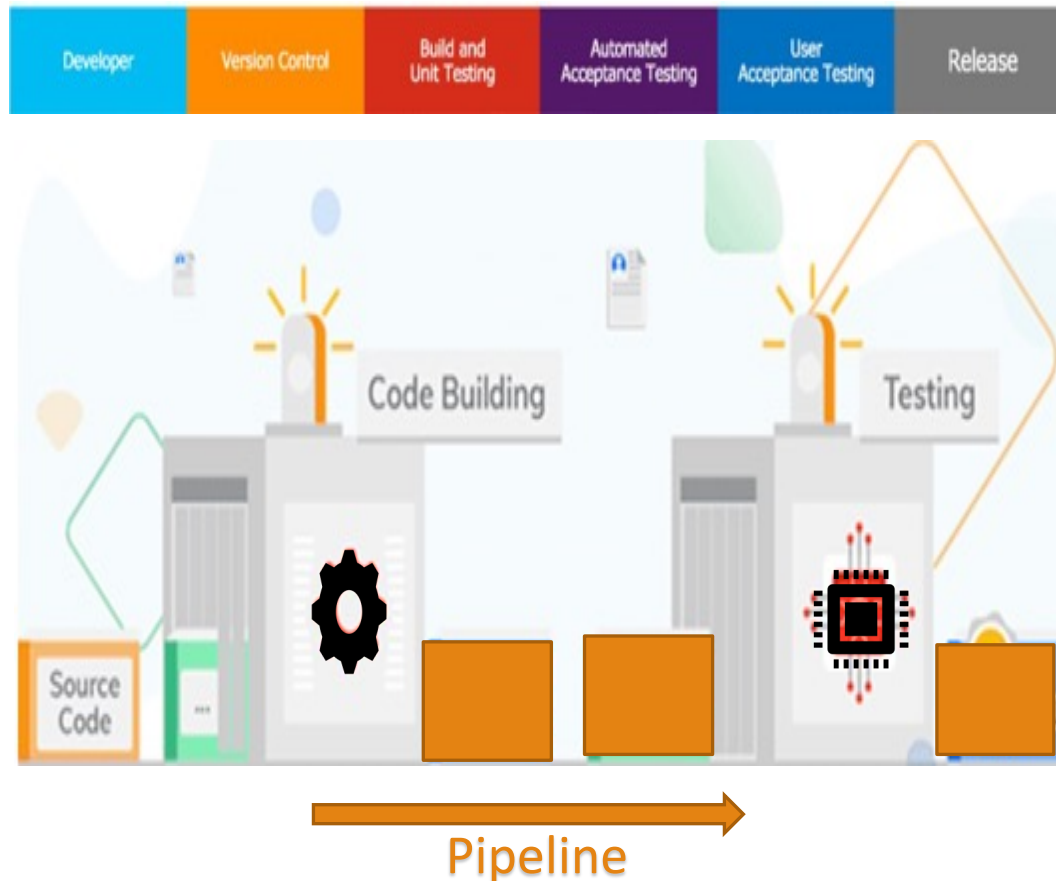
Manual & Automation Testing

- **Manual testing** is the process in which QA analysts execute tests one-by-one in an individual manner. The purpose of manual testing is to catch bugs and feature issues before a software application goes live.
- **Automation testing** is the process in which testers utilize tools and scripts to automate testing efforts.
- In manual testing, a **human** performs the tests step by step, without test scripts.
- In automated testing, tests are executed automatically via test automation frameworks, along with other **tools** and software.
- Testers should learn the skills of Agile and DevOps methodologies. The Agile method of working provides speed to the test project.

Implementing Automated Testing for Agile & Traditional Methodology



Implementing DevOps Automated Testing Continuous Integration & Continuous Delivery



Summary

- Software Testing Principles – **Early Testing** (Save Money), **Detect Cluster together** (Error in same area), **Pesticide paradox** (repeat similar ineffective to find errors) and **testing is context dependent**.
- Test Method – **Black box** Testing (unknown internal implementation), **White box** Testing (known internal implementation) and **Grey box** Testing (combine both)
- Testing Framework - **Test-Driven Development** (focus on implementation of a features and writing unit test case), **Behavioral Driven Development** (focus on creating simple scenarios on how an application should behave) & **Acceptance Test-Driven Development** (focus on capturing user requirement criteria to tests)

Summary

- Static Testing Techniques – **Informal Reviews** (display content to seek opinion of everyone), **Walk Through** (perform checks by expert or experience person), **Peer Review**(seek another colleague review), **Inspection**(verification of document) and **Static Analysis** (evaluation of code)
- Functional Testing verifies that each function of the software application operates in conformance with the requirement specification
- Dynamic Testing Techniques (Function) – **Unit Testing** (test individual component), **Integration Testing** (test two or more module together), **System Testing** (complete/entire software testing) and **User Acceptance Testing** (for delivery to customer or going live).

Summary

- Non-Function Testing determine the performance of the system and to validate or verify the quality attribute of the system
- Dynamic Testing Techniques (Non Function) – **Performance Testing** (speed, response time, stability, reliability, scalability and resource usage), **Security Testing** (uncovers vulnerabilities, threats, risks and prevent malicious attack) and **compatibility Testing** (testing to check whether your software is able of operating on different environment)
- Type of Implemented Testing – **Manual & Automated Testing**

References:

Testing framework:

- <https://phoenixnap.com/blog/tdd-vs-bdd>
- <https://www.softwaretestingmaterial.com/tdd-vs-bdd-vs-atdd>

Testing type, technique & Tactics:

- <https://tutorialshouse1.wordpress.com/2013/06/26/testing-methodologies>
- <https://www.janio.careers/post/grey-box-testing-with-a-janio-qa-engineer>

Measurement in software testing -

<https://www.vskills.in/certification/tutorial/measurement-in-software-testing/>

Related Process

- <https://www.guru99.com/what-everybody-ought-to-know-about-test-planing.html>
- <https://www.panaya.com/blog/testing/what-is-uat-testing>
- <https://qatestlab.com/resources/knowledge-center/5-types-of-test-data-that-should-be-used-during-software-testing/>